SOLAR: Complete Proofs

Extended Supplement to "When Classic Cache Policies Fail:
Learning-Augmented Replacement for Semantic Retrieval Buffers"

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This supplement contains the complete proofs of the theorems in Section 5 ("Theoretical Analysis") of the main paper. The main paper provides proof sketches sufficient to verify each claim; here we restate each result and give the full proof. Theorem numbering follows the order of presentation in the main paper. This document is restricted to mathematical proofs and does not reproduce the method, system design, or experimental content of the main paper.

Notation. K is the cache capacity, T the horizon, k the retrieval depth, $\lambda > 0$ the per-modification switching cost, and τ the admission threshold. The per-step miss cost is $\ell_t \in [0, 1]$ with $\delta_t = 1 - \text{qual}_t$. OPT denotes the offline optimal policy.

1 FIFO Has Unbounded Competitive Ratio

Theorem 1 (FIFO CR is $\Omega(K)$)**.** *For cache size K and any constant $M > 0$, there exists an input sequence σ such that* $\text{COST}(\text{FIFO}, \sigma)/\text{COST}(\text{OPT}, \sigma) \geq M$. *Specifically,* $\text{CR}(\text{FIFO}) \geq K + 1$ *in the soft-hit semantic cache model.*

Proof. We construct a family of adversarial sequences parametrized by $M > 0$.

Construction. Let the embedding space be $\mathbb{R}^d$ with $d \geq m$. Define $m = K + \lceil M \rceil$ topic centers $\{c_1, \ldots, c_m\}$ as orthonormal vectors (achievable when $d \geq m$). Each topic j has a single item e_j at position c_j and generates queries q near c_j (with $\text{sim}(q, e_j) > 0.9$ and $\text{sim}(q, e_i) < 0.1$ for $i \neq j$).

The query sequence is a strict cycle: $q_1 \in T_1, q_2 \in T_2, \ldots, q_m \in T_m, q_{m+1} \in T_1, \ldots$ At each step t, a new item e_t from the current topic is generated.

FIFO behavior. After the initial fill (steps $1, \ldots, K$), the cache contains items from topics $T_1, \ldots, T_K$. At step $K + 1$ the query targets T_{K+1}, which is not cached (miss); the new item is admitted and FIFO evicts the oldest item (from T_1), leaving $T_2, \ldots, T_{K+1}$. At step $K+2$ the query targets T_1 (just evicted, miss), and FIFO evicts T_2. By induction, at every step $t > K$ the query targets a topic whose item was evicted at step $t - K$, so every step after fill is a complete miss ($\ell_t = 1$).

FIFO's total miss cost is $(T - K) \cdot 1$ and its switching cost is $(T - K) \cdot \lambda$ (one eviction per step after fill), giving

$$\text{COST}(\text{FIFO}) = (T - K)(1 + \lambda). \tag{1}$$

OPT behavior. OPT retains K topics permanently (no evictions after the initial fill). Because all $m = K + 1$ topics are equiprobable in the cycle, retaining any fixed K of them is optimal: any additional replacement only adds switching cost without reducing the per-cycle miss count. OPT hits on K of every $K + 1$ queries, missing on one, so

$$\text{COST}(\text{OPT}) = \frac{T}{K + 1} + \lambda K. \tag{2}$$

Ratio. The initial fill contributes $O(K)$ to both costs and is dominated as $T \to \infty$. Hence

$$\mathrm{CR}(\mathrm{FIFO}) \geq \lim_{T\to\infty} \frac{(T-K)(1+\lambda)}{T/(K+1)+\lambda K} = (K+1)(1+\lambda). \tag{3}$$

Since K is a system parameter that can be made arbitrarily large, the ratio is unbounded: for any M, setting $K \geq M$ yields $\mathrm{CR} \geq M$. (The bound is in fact $(K+1)(1+\lambda)$; we state the weaker $\Omega(K)$ form because the $(1+\lambda)$ factor is immaterial to the unboundedness conclusion.) For an even stronger result, set $m = T^{1/2}$ and $K = 1$: $\mathrm{CR} \geq (T-1)(1+\lambda)/[T(T^{1/2}-1)/T^{1/2}+\lambda] \approx T^{1/2} \to \infty$. □

2 SOLAR Bounds the Competitive Ratio

Theorem 2 (SOLAR Replacement Bound)**.** *Under cost-aware admission with thresholds* $\{\tau_t\}$ *and* $\tau_{\min} = \min_t \tau_t$*:* $N^{\mathrm{SOLAR}} \leq \lfloor T/\tau_{\min} \rfloor$.

Proof. An admission occurs at time t only when the accumulated cost c since the last admission satisfies $c \geq \tau_t$; after each admission, c is reset to 0. Since $\delta_t = 1 - \mathrm{qual}_t \leq 1$, re-accumulating to the active threshold (which is $\geq \tau_{\min}$) requires at least $\lceil \tau_{\min} \rceil$ steps. Hence the inter-admission gap is $\geq \tau_{\min}$, and over T steps

$$N^{\mathrm{SOLAR}} \leq \lfloor T/\lceil \tau_{\min} \rceil \rfloor \leq \lfloor T/\tau_{\min} \rfloor. \tag{4}$$

With a fixed threshold τ this is $\lfloor T/\tau \rfloor$. Under the EMA adaptation, τ_t is clipped below by a floor $\tau_{\mathrm{floor}} > 0$, so $\tau_{\min}$ equals the converged value up to a short initial transient. □

Definition 1 (Bounded stale advantage)**.** *Assume single-item bounded influence: swapping one cached item changes the per-step miss cost by at most* $1/k$*, where* k *is the retrieval depth. A workload has* bounded stale advantage *if, in any epoch during which* OPT *holds a fixed cache,* OPT*'s accumulated miss cost over that epoch is at least half of SOLAR's accumulated miss cost over the same epoch.*

Theorem 3 (SOLAR Competitive Ratio)**.** *Under Definition 1, with switching cost* $\lambda > 0$ *and the fixed threshold* $\tau = 2\lambda$*,* $\mathrm{CR}(\mathrm{SOLAR}) \leq 3$*, a constant independent of* K*,* T*, and* λ*.*

Proof. Partition $[1, T]$ into epochs $E_j = [t_j, t_{j+1})$ delimited by SOLAR's admission times.

SOLAR cost per epoch. Within an epoch SOLAR's cache is fixed and changes exactly once, at the end. By the threshold trigger the accumulated miss cost is τ (the final epoch contributes $\leq \tau$), and there is one admission with switching cost λ. Hence $\mathrm{COST}(\mathrm{SOLAR}, E_j) \leq \tau + \lambda$.

OPT cost per epoch. Two cases.

Case 1: OPT replaces at least once in E_j*.* Then OPT incurs switching cost $\geq \lambda$.

Case 2: OPT holds a fixed cache S^* *throughout* E_j*.* Let the two caches differ in D_j items. Under single-item bounded influence, the per-step miss costs of the two fixed caches differ by at most D_j/k, so over the epoch of length $|E_j|$,

$$\sum_{t\in E_j} \ell_t(S^*) \;\geq\; \sum_{t\in E_j} \ell_t(S^{\mathrm{SOLAR}}) - \frac{|E_j|\, D_j}{k} \;=\; \tau - \frac{|E_j| D_j}{k}. \tag{5}$$

The bounded-stale-advantage condition (Definition 1) is precisely $|E_j| D_j/k \leq \tau/2$, hence $\mathrm{COST}(\mathrm{OPT}, E_j) \geq \tau/2$.

Combining. In both cases $\text{COST}(\text{OPT}, E_j) \geq \min(\lambda, \tau/2)$. Additional replacements by OPT within an epoch only increase its cost, so no charging across epoch boundaries is needed. Summing,

$$\text{CR} \leq \frac{\tau + \lambda}{\min(\lambda,\, \tau/2)}. \tag{6}$$

Optimal threshold. The bound (6) is minimized when the two arguments of the min coincide. For $\tau \leq 2\lambda$ it equals $2 + 2\lambda/\tau$ (decreasing in τ); for $\tau \geq 2\lambda$ it equals $1 + \tau/\lambda$ (increasing in τ). The minimum is at $\tau = 2\lambda$, where both equal 3:

$$\text{CR} \leq \frac{2\lambda + \lambda}{\lambda} = 3. \tag{7}$$

The CR-optimal threshold $\tau = 2\lambda$ differs from the cost-optimal $\tau^\star = \sqrt{2\lambda/L}$ (which minimizes SOLAR's own average cost for loss rate L, not the worst-case ratio); the two coincide when $L = 1/(2\lambda)$. □

Stationary convergence. Under a piecewise-stationary workload with P phase changes, each stationary segment has a fixed optimal cache. SOLAR's adaptive threshold τ_t converges to $\tau_i^\star = \sqrt{2\lambda/L_i}$ within $O(1/\alpha)$ steps of entering phase i (with L_i the stationary loss rate). The excess cost over OPT per phase is bounded by the convergence transient $O(\sqrt{\lambda/L_i})$, so $\text{COST}(\text{SOLAR}) - \text{COST}(\text{OPT}) \leq P \cdot O(\sqrt{\lambda}) + P\lambda$, giving $\text{CR} = 1 + P \cdot O(\sqrt{\lambda})/(T\bar{L}) \to 1$ as $T \to \infty$ for fixed P.

3 Regret Bounds

Theorem 4 (FIFO Linear Regret)**.** *On cycling workloads with $m > K$, $R_T(\text{FIFO}) = \Omega(T)$.*

Proof. On the cycling workload of Theorem 1 with $m = K + 1$, FIFO achieves hit rate 0 after fill: $\sum_{t=K+1}^{T} \ell_t = T - K$. We compare against the best fixed-eviction policy on *the same admit-all sequence.* Since FIFO admits every item, a competing eviction policy may immediately evict each newly admitted item and thereby retain the first K topics permanently, achieving miss rate $1/(K + 1)$, i.e. $\sum_t \ell_t^* = T/(K + 1)$. Therefore

$$R_T = (T - K) - \frac{T}{K+1} = T \cdot \frac{K}{K+1} - K = \Omega(T). \tag{8}$$

FIFO's per-step loss never improves regardless of interaction length. □

Theorem 5 (Thompson Sampling Eviction Regret)**.** *Under stochastic item utilities with K cache slots, $R_T(\text{TS}) \leq O(\sqrt{KT \log T})$.*

Proof. We analyze eviction under a stylized stationary model whose assumptions we state explicitly. Fix the pool of items competing for cache slots and assume each item i has a stationary latent utility μ_i, with retrieval feedback providing a Bernoulli observation of whether i was useful for the current query. Map eviction to a K-armed bandit: arms are the cached items, "pulling" arm i corresponds to retaining i and observing its feedback, and the regret target is identifying the lowest-utility item to evict.

Under this model the standard Beta-Bernoulli Thompson Sampling analysis [1] gives per-arm regret $O(\log T/\Delta_i)$, where Δ_i is the utility gap to the worst item. The worst-case gap instantiation $\Delta_i = \Theta(\sqrt{K/T})$ for all suboptimal arms yields total regret $O(\sqrt{KT \log T})$.

Scope of the reduction. Two modeling gaps separate this idealization from SOLAR's exact dynamics. (i) The arm set is not literally fixed: each admission introduces a new item and removes one, so the bandit instance drifts slowly. (ii) The reward (improvement in cache quality) is observed only indirectly through subsequent retrievals rather than immediately. SOLAR's aging mechanism (β_i += 0.05 every 5 steps for un-retrieved items) limits the effective observation window, so within a window the instance is approximately stationary and the bound applies windowwise. A fully non-stationary treatment (sliding-window or restless-bandit regret) would replace $\log T$ by a window-dependent factor while preserving the dominant $\sqrt{KT}$ scaling. □

Theorem 6 (Lower Bound)**.** *For any online eviction policy, $R_T(\pi) \geq \Omega(\sqrt{KT})$.*

Proof. We invoke the *minimax* (worst-case) multi-armed bandit lower bound [2], which is $\Omega(\sqrt{KT})$ *without* a logarithmic factor (in contrast to the gap-dependent bound $\Omega(K \log T/\Delta)$). Construct K items where one item i^* has utility $1/2 - \epsilon$ and the rest $1/2$; the optimal eviction always targets i^*. Any policy must spend $\Omega(1/\epsilon^2)$ observations per suboptimal arm to identify i^*. Setting $\epsilon = \Theta(\sqrt{K/T})$ over the $K - 1$ suboptimal arms gives

$$R_T \geq (K-1) \cdot \epsilon \cdot \frac{T}{K} = \Omega(\sqrt{KT}). \tag{9}$$

This applies to any online eviction policy, so the Thompson Sampling upper bound $O(\sqrt{KT \log T})$ is optimal up to the $\sqrt{\log T}$ factor. □

References

[1] Shipra Agrawal and Navin Goyal. Analysis of thompson sampling for the multi-armed bandit problem. In Shie Mannor, Nathan Srebro, and Robert C. Williamson, editors, *Proceedings of the 25th Annual Conference on Learning Theory*, volume 23 of *Proceedings of Machine Learning Research*, pages 39.1–39.26, Edinburgh, Scotland, 25–27 Jun 2012. PMLR.

[2] Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.